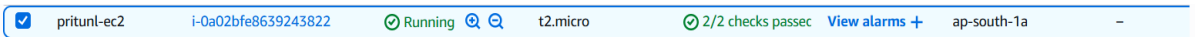
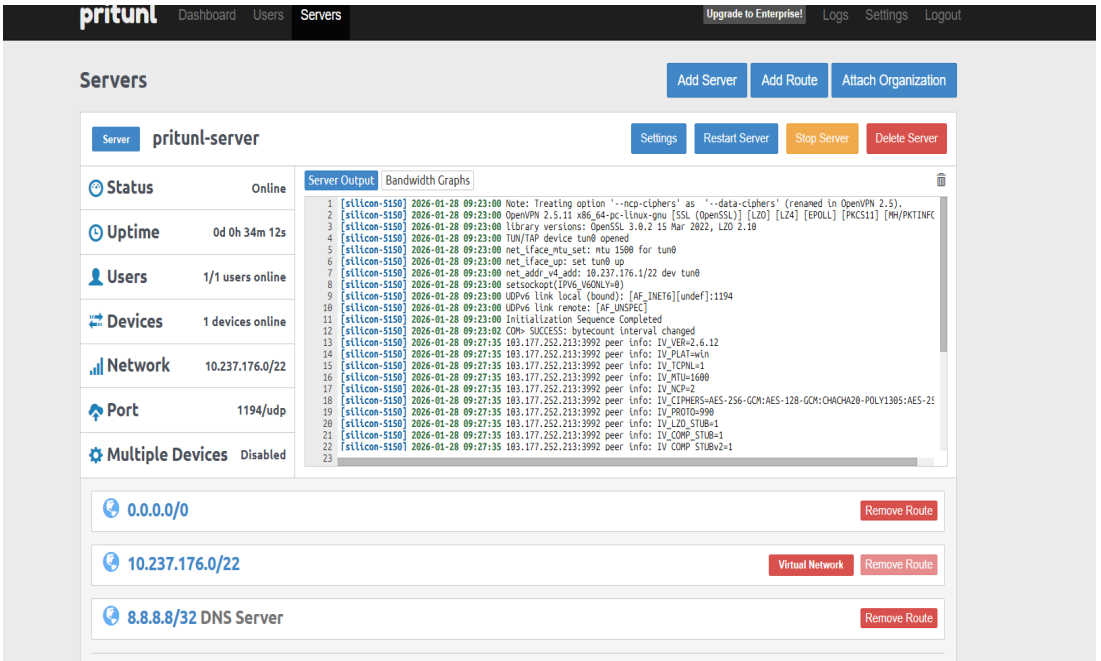
Postgres Connect through Pritunl VPN

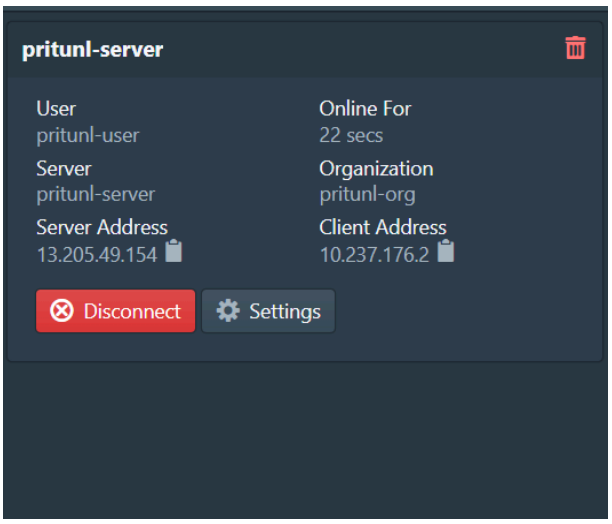
- This setup enables a secure connection to a private PostgreSQL RDS instance via a Pritunl VPN server running on a public EC2 instance.
- Clients can connect to RDS only through VPN.
- Public instance runs Pritunl VPN. VPN client gets IP from Pritunl VPN network.
- Created a VPC with a public subnet (igw) and two private subnets for RDS subnet groups with NAT gateway which has the public subnet as source for external connectivity.
- Created an EC2 instance instance with ubuntu.



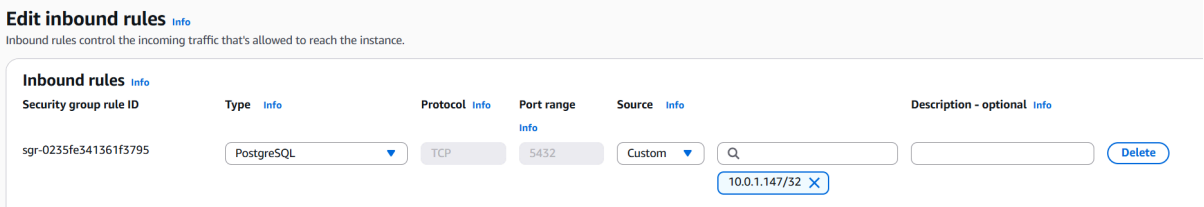
- Installed Pritunl VPN and mongodb service inside the instance.
- Started and enabled the service. Accessed pritunl UI via the public IP of the instance
- Created Organisation and User to attach in a created Server to start the server.
- Added the port specified in pritunl server in the inbound rules in security group for the pritunl EC2.
- Downloaded the profile of the user as .tar file and imported the profile in the pritunl client running on the local system.



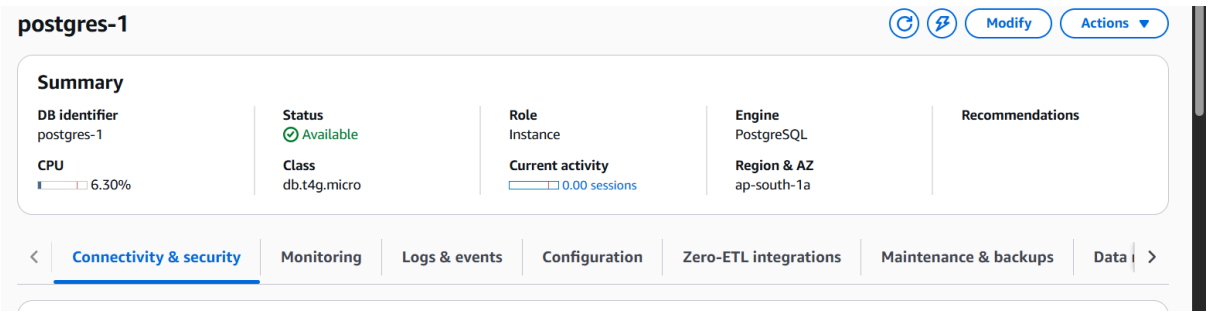
- Connected to the user and in the Pritunl UI it was seen that the user was online.



- Created a subnet group for RDS and provided the two private subnets that were created.
- Created security group for RDS with inbound traffic from 5432 (postgresSQL) and provided the private IP of the pritunl EC2 as the souce and set all traffic as outbound.



- VPN clients connect via EC2 public IP and RDS traffic is NATed through EC2 private IP. NAT ensures RDS only sees EC2 private IP.
- Created RDS instance with PostgreSQL as engine.in free tier

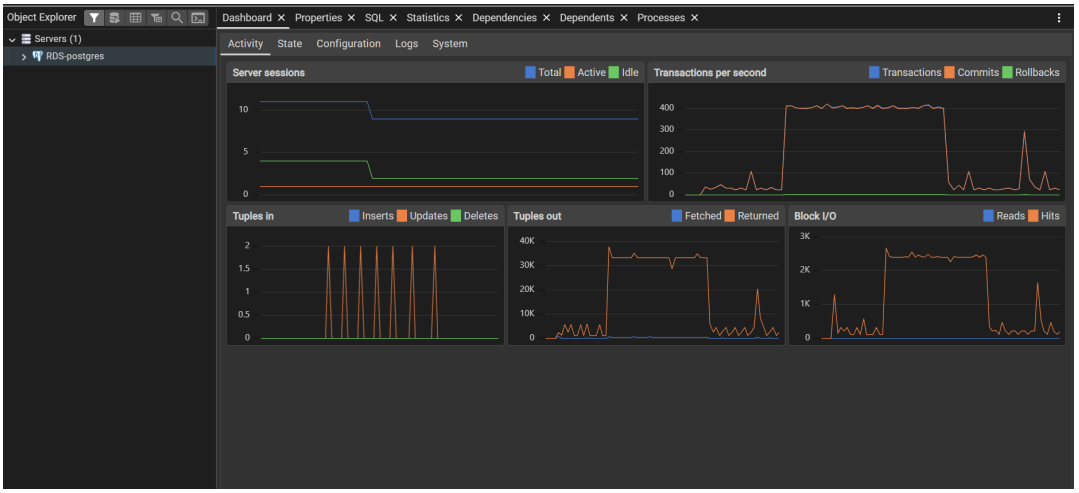
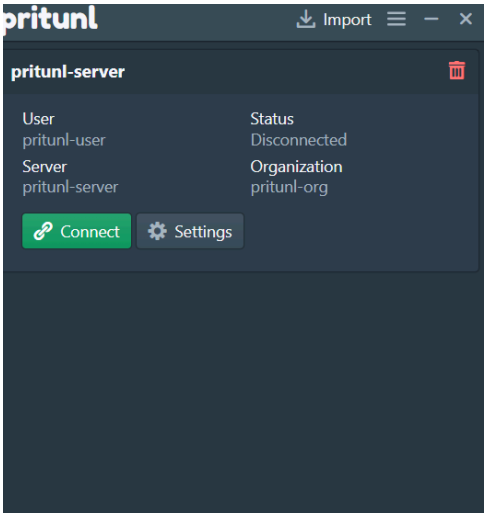
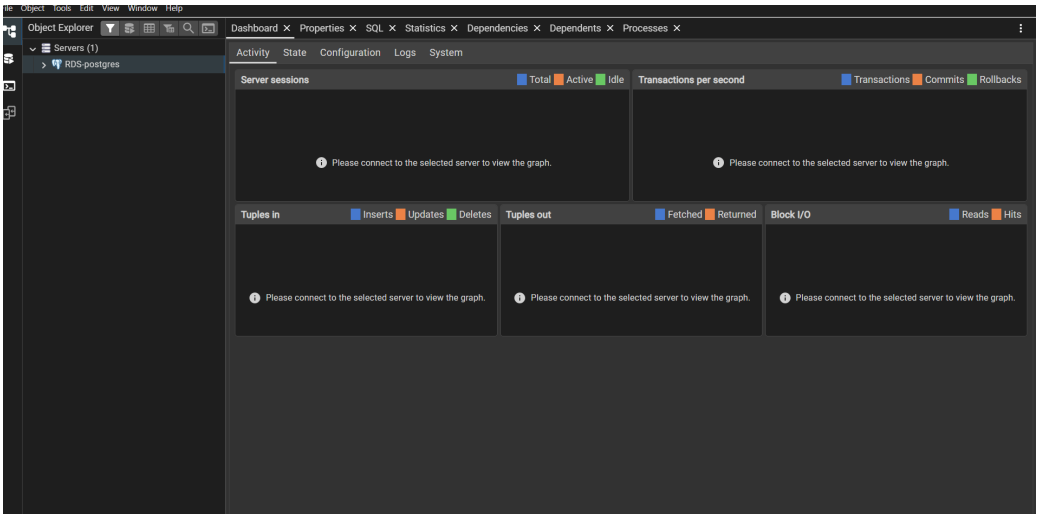


- Defined the database name, master username and also generated the master password.
- Specified the created subnet, subnet group, security group for rds instance and created the instance.
- Obtained the endpoint for the RDS and tested its connectivity locally inside the EC2 instance.

```
ubuntu@ip-10-0-1-147:~$ nc -vz postgres-1.cze44o6c4du9.ap-south-1.rds.amazonaws.com 5432
Connection to postgres-1.cze44o6c4du9.ap-south-1.rds.amazonaws.com (10.0.2.208) 5432 port [tcp/postgresql] succeeded!
```

- Opened PgAdmin and registered a new server.

- Specified the RDS endpoint, database name, username and password in the connection tab of the server.
- In parameters provided SSL mode required. Saved the configuration and the databse was successfully connected.



- The PostgresSQL stays connected as long as the VPN client is active.
- As soon as the Pritunl VPN client is disconnected the database also gets disconnected.